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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

ME1312-1 – INTRODUCTION TO MACHINE LEARNING

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. We need machine learning for the following reasons –
 - (i) Tasks which are difficult to program
 - (ii) Tasks which are beyond human capabilities
 - (iii) Ability to adapt
 - (iv) Ability to survive

A) (i), (ii) and (iii) B) (i) and (ii)
C) (ii), (iii) and (iv) D) (iii) and (iv)
2.path is the chain of transformations from input to output
A) Critical B) Credit assignment
C) Processing D) Universal
3. High-bias machine learning algorithm
A) Logistic regression B) Decision trees
C) Support vector machines D) k-means clustering
4. In the sigmoid activation function, the synaptic weights are changed the most for those neurons in the network, where the signals are in their mid-range and this contributes to the back propagation learning algorithm
A) plasticity B) Stability
C) equilibrium D) Balance
5. The design of a neural network as a 'curve fitting (approximation) problem' in a high dimensional space is the principle of
A) Multilayer perceptron B) Radial basis function network
C) Convolutional network D) Lattice network
6. Which of the following statements is not true about the Decision tree?
A) It can be applied on binary classification problems only. B) It is a predictor that predicts the label associated with an instance by traveling from a root node of a tree to a leaf.
C) At each node, the successor child is chosen on the basis of a splitting of the input space. D) The splitting is based on one of the features or on a predefined set of splitting rules
7. The theoretical analysis of MLP becomes difficult due to the distributed form of..... and high degree of
A) Nonlinearity and connectivity B) Complexity and bias
C) Connectivity and nonlinearity D) Variance and bias
8. Pattern-by-pattern updating of weights makes the search in weight space in nature, because of the algorithm is not stuck in local minima.
A) continuous B) Stochastic
C) Linear D) Redundant

9.network is used to handle time series data.
 A) feedforward B) Radial basis function
 C) convolutional D) Recurrent
10. Kernel type functions used in RBFNN suffer from
 A) overfitting B) Curse of dimensionality
 C) underfitting D) Memorization
11. In the context of input-output mapping, the selection of the 'simplest' function in the absence of any prior knowledge is called
 A) Cover's theorem B) Curse of dimensionality
 C) Vapnik-Chervonenkis dimension D) Occam's razor
12.type of learning uses 'labelled' training data.
 A) Semi-supervised B) Unsupervised
 C) Reinforcement D) Supervised
13. error provides a measure of how much performance is lost as a result of using a training sample of some prescribed size N.
 A) Approximation B) Mean square
 C) absolute D) Estimation
14. The following is not a technique used in learning artificial neural networks
 A) Supervised learning B) Random learning
 C) Reinforcement learning D) Competitive learning
15. The nonlinearity used in the activation functions of MLP should be
 A) integrable B) Differentiable
 C) distributed D) Local
16. Radial basis function neural network uses for calculating the output
 A) Dual norm B) Chebyshev norm
 C) Euclidean norm D) Absolute norm
17. The parameter tells how much support vector machine optimization can avoid misclassifying each training example
 A) kernel B) minimization
 C) Regularization D) error
18. Non-linear SVM uses kernel functions to
 A) To make non-separable data into separable B) To map data into a low dimensional space
 C) To make separable data into non-separable D) To map data into a high dimensional space
19.purpose of zero-padding in a CNN.
 A) To adjust the learning rate during training B) To prevent the reduction of spatial dimensions
 C) To regularize the network and prevent overfitting D) To improve the sampling process
20. The following are the real-world applications of Support vector machines.....
 A) Text and hypertext classification B) Image classification
 C) All of the above D) Clustering of the news articles

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Write a note on Artificial intelligence. Differentiate between artificial intelligence and machine learning.	05	L*2	01	01
	b) Discuss the following learning tasks – pattern association and pattern recognition.	08	L3	02	01
	c) Write a note on cumulative distribution function.	03	L2	01	01
2.	a) Discuss the different machine learning algorithms.	08	L2	01	01
	b) What are the different types of activation functions used in neural networks?	05	L2	02	01

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|----|--|----|----|----|----|
| c) | What are the characteristics of supervised learning? | 03 | L2 | 02 | 01 |
|----|--|----|----|----|----|
3. a) Write a note on logistic regression. How is it different from linear regression? 08 L3 01 01
- b) How is Hebbian learning different from other learning methods? 05 L3 02 01
- c) How do you relate a neuron with an artificial neuron? 03 L2 02 01

Unit – II

4. a)

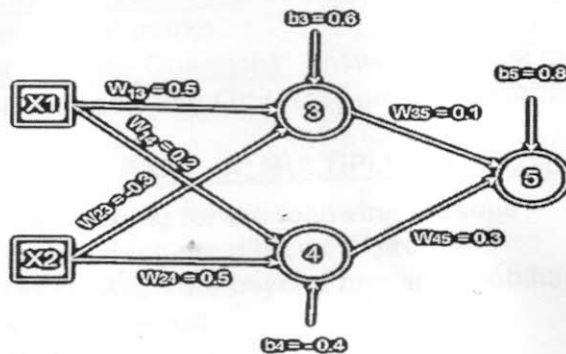


Figure. 4a

Consider a MLPNN as shown in figure. 4a. The initial weights and biases are as given in the figure. Taking a learning rate of 0.5 and considering the input-output training data as follows – (1, 1, 0) and (0, 1, 1), update the w_{35} , w_{45} and b_5 using back-propagation algorithm. Use sigmoid activation function in neurons 3, 4 and 5.

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|----|----------------------------------|----|----|----|----|
| b) | What is Curse of dimensionality? | 14 | L3 | 03 | 01 |
| | | 02 | L2 | 04 | 01 |
5. a) Discuss early stopping method of training in MLP. 06 L2 03 01
- b) Discuss the receptive field concept in the context of RBFNN. 10 L2 04 01
6. a) Explain sigmoidal nonlinearity and its use in error back propagation algorithm. 08 L2 03 01
- b) Explain the structure of a radial basis function neural network. How is Cover's theorem applicable to working of RBFNN? 08 L3 04 01

Unit-III

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|-------|---|----|----|----|----|
| 7. a) | Discuss Quadratic optimization for finding the optimal hyperplane in SVM. | 11 | L3 | 05 | 01 |
| b) | What are the disadvantages of support vector machines? | 05 | L2 | 05 | 01 |
8. a) What is Mercer's theorem? 05 L2 05 01
- b) Write a note on convolutional neural network. 05 L2 05 01
- c) Compare SVM with RVM. 06 L2 05 01

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
